StackRec Report: Progressive Layer Stacking for SASRec

Experiments on Beauty Dataset (reviews_Beauty_5) — October 2026

1. Introduction

StackRec is a progressive depth-narrowing method adapted for the SASRec (Sequential Recommendation) model. Instead of training a deep model from scratch, training starts with the bottom 2 transformer layers, then weights are copied to new layers and all 6 layers are gradually activated.

Key property: when layer scale=0, lower layers act as identity functions, so a 2-layer subnet inside a 6-layer model produces **identical** predictions to a standalone 2-layer SASRec (tests 16-17, 17/17 PASS).

2. StackRec Architecture

The model always has 6 transformer layers. Per-layer forward coefficients (*layer_scales*) control layer activation: scale=0 means identity (skip), scale=1 means full activation.

Training stages:

- **Stage 1 (2_layers, 90 epochs):** train layers 4,5; layers 0-3 frozen with scale=0.
- **Stage 2 (4_layers, 60 epochs, 10-epoch ramp):** copy weights 4->2, 5->3; train layers 2-5; scales of layers 2,3 ramp from 0 to 1.
- **Stage 3 (6_layers, 60 epochs, 10-epoch ramp):** copy 2->0, 3->1; train all 6 layers; scales of layers 0,1 ramp from 0 to 1.

Each stage uses a smaller LR: $lr(L) = lr_ref \times sqrt(4/L)$.

3. Experiments

3.1. Baseline SASRec (AdamW / Muon, L=2,4,6)

Increasing depth degrades performance for both optimizers. AdamW consistently outperforms Muon.

Optimizer	L=2	L=4	L=6
AdamW	0.012762	0.004797	0.005645
Muon	0.005187	0.003540	0.003692

3.2. AdamW + StackRec

StackRec achieves 2.3-2.7x higher test NDCG@10 for 4 and 6 layers compared to plain AdamW.

Layers	LR	Epoch	Val NDCG@10	Test NDCG@10	Test Rec@10
2 (4,5)	0.00297	88	0.014714	0.011958	0.023380
4 (2-5)	0.00210	129	0.015451	0.012964	0.025370

6 (all)	0.00171	194	0.016095	0.013299	0.025494

3.3. Muon + StackRec

Muon + StackRec improves over plain Muon for 4 and 6 layers, but remains significantly below AdamW + StackRec. Best Muon stage: 4 layers.

Layers	Muon LR	AdamW LR	Epoch	Val NDCG@10	Test NDCG@10
2	0.02828	0.00297	85	0.007215	0.004602
4	0.02000	0.00210	119	0.007884	0.004518
6	0.01633	0.00171	173	0.007008	0.004341

3.4. Auxiliary Identity Loss

When a new 2-layer group is stacked below the trained subnet, an auxiliary loss drives the output ratio to 1: $\lambda(alpha) = \lambda_0 \times (1 - alpha)$.

Opt.	Layers	Val NDCG@10	Test NDCG@10
AdamW	2	0.014714 / 0.014714 (+0.00%)	0.011958 / 0.011958 (+0.00%)
AdamW	4	0.015451 / 0.015920 (+3.04%)	0.012964 / 0.012934 (-0.23%)
AdamW	6	0.016095 / 0.016328 (+1.45%)	0.013299 / 0.013086 (-1.60%)
Muon	2	0.007215 / 0.007215 (+0.00%)	0.004602 / 0.004602 (+0.00%)
Muon	4	0.007884 / 0.007327 (-7.06%)	0.004518 / 0.004591 (+1.61%)
Muon	6	0.007008 / 0.006883 (-1.79%)	0.004341 / 0.004351 (+0.24%)

3.5. Hyperbolic Heads

Poincare-ball next-item head at each newly stacked group boundary. Curvature decays with depth: $c(d) = c_ref \times (d_ref / d)^D$.

Layers	Plain	Aux	Hyperbolic
2	0.011958	0.011958	0.011958
4	0.012964	0.012934	0.012428
6	0.013299	0.013086	0.013419

6-layer model gains +0.9% vs plain, +2.5% vs aux. 4-layer stage is weaker -- loss_coeff=0.1 may be too small.

3.6. Lorentz Hyperbolic Residual

Hyperbolic geometry *inside* the hidden representation of copied groups: $h_out = normalize(h + gate \times normalize_to_norm(h, r))$, where r = (sinhc(sqrt(c)·||u||) - 1) × W_out u is the curvature-induced correction,

exactly zero at c=0.

Verification: unit-level checks (CPU) -- hyperboloid constraint, symmetric distance, zero residual at c=0, bitwise identity at gate=0, norm preservation. Full training not yet run.

4. Testing

4.1. Ranking Metrics Tests (10/10 PASS)

#	Metric	Status	Value
1	NDCG@5	PASS	0.487914
2	Recall@5	PASS	0.800000
3	Coverage@5	PASS	1.000000
4	Novelty@5	PASS	0.944260
5	MeanRank@5	PASS	3.500000
6	ILD@5	PASS	0.506127
7	PopularityGroups	PASS	OK
8	EdgeCases	PASS	OK
9	CoveragePartial	PASS	0.100000
10	MeanRankPartial	PASS	1.000000

Notes: NDCGMetric computes DCG (no IDCG normalization). RecallMetric counts absolute hits. CoverageMetric and NoveltyMetric are stateful: require .reduce().

4.2. Temporal Split Validation

Split	Rows	Users	Items
train	140,965	18,474	11,534
validation	50,336	4,725	10,310
test	90,621	8,041	11,727
train_validation	151,081	18,474	11,736

Temporal split violations: **0**. Prefix property *train* $\subset$ *val* $\subset$ *test*: **True**.

4.3. StackRec Unit Tests (17/17 PASS)

1	Model initialization	PASS
2	Forward pass (short seq=5)	PASS
3	Forward pass (long seq=20)	PASS
4	Stage 1 -- 2 layers (4, 5)	PASS
5	Stage 2 -- 4 layers (2-5) with copy	PASS
6	Stage 3 -- 6 layers (0-5) with copy	PASS
7	Full StackRec pipeline (short seq=5)	PASS
8	Full StackRec pipeline (long seq=20)	PASS
9	Ramp scale=0 identity	PASS
10	Aux scalar and boundary	PASS

11	Lorentz head addition	PASS
12	Curvature schedule	PASS
13	Copy layer weights	PASS
14	Batch with two sequences	PASS
15	CPU only	PASS
16	SASRec vs StackRec (2 layers) identity	PASS
17	SASRec vs StackRec (2 layers) long seq	PASS

Key checks: Ramp scale=0 -- new layers as identity; SASRec vs StackRec identity -- identical predictions; Copy layer weights -- correct copying and trainability switching.

5. Conclusions

• **StackRec is effective:** AdamW + StackRec achieves 2.3-2.7x higher test NDCG@10 for 4 and 6 layers vs. plain AdamW.

• **Muon + StackRec:** Improves over plain Muon for 4 and 6 layers, but remains below AdamW. Best stage: 4 layers.

• **Auxiliary identity loss:** Improves validation for AdamW, but test NDCG slightly worse. For Muon: mixed/negative results.

• **Hyperbolic heads:** Depth-decaying curvature helps at 6 layers (+0.9% vs plain), but 4-layer stage is weaker.

• **Lorentz residual:** Verified at unit level; full training not yet run.

• **Testing:** 10/10 metrics PASS, 17/17 StackRec tests PASS.

Project files: sasrec_experiments/

Sources: diary.md, test_stackrec.py, test_metrics.py, sasrec_stackrec_results.json, sasrec_stackrec_muon_results.json, sasrec_stackrec_aux_results.json, sasrec_stackrec_hyperbolic_results.json